



SMIE 2017-1 - A sample paper to prepare for the final examination.

Statistical Modelling and Inference II (The University of Adelaide)



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Examination in School of Mathematical Sciences
Semester 2, 2017

104843 STATS 2107 Statistical Modelling & Inference II

Official Reading Time: 10 mins
Writing Time: 120 mins
Total Duration: 130 mins

NUMBER OF QUESTIONS: 5 TOTAL MARKS: 70

Instructions

- Attempt all questions.
- Begin each answer on a new page.
- Examination materials must not be removed from the examination room.

Materials

- 1 Blue book is provided.
- Calculators without remote communications capability are allowed.
- English and foreign-language dictionaries may be used.

DO NOT COMMENCE WRITING UNTIL INSTRUCTED TO DO SO.

1. Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed (*i.i.d.*) random variables with probability density function $f(y; \theta)$ for a real scalar parameter $\theta \in \Theta$, where Θ denotes the parameter space. Let $T = T(Y_1, Y_2, \dots, Y_n)$ be an estimator for θ .

(a) Define the mean squared error, $MSE_T(\theta)$, of T .

(b) Define the bias, $b_T(\theta)$, of T .

(c) Prove that

$$MSE_T(\theta) = \text{var}(T) + b_T(\theta)^2.$$

- (d) Suppose $Y_1, Y_2, \dots, Y_n$ are *i.i.d.* Bernoulli random variables with probability of success $0 \leq p \leq 1$, and that $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ is to be used as an estimator for p .

(i) Show that $\bar{Y}$ is an unbiased estimator for p .

(ii) Calculate $MSE_{\bar{Y}}(p)$.

[11 marks]

2. Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed (*i.i.d.*) random variables with probability density function $f(y; \theta)$ for a real scalar parameter $\theta \in \Theta$, where Θ denotes the parameter space.

(a) Define a $100(1 - \alpha)\%$ confidence interval for the parameter θ .

(b) Suppose that $Y_1, Y_2, \dots, Y_n$ are *i.i.d.* $N(\mu, \sigma^2)$. Let c_1, c_2 be such that

$$P(c_1 < X < c_2) = 1 - \alpha,$$

where

$$X \sim \chi_{n-1}^2.$$

(i) Prove that the interval

$$\left(\frac{(n-1)S^2}{c_2}, \frac{(n-1)S^2}{c_1} \right),$$

where S^2 is the sample variance, is a $100(1 - \alpha)\%$ confidence interval for σ^2 . You may assume that

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2.$$

- (ii) In an experiment, 20 observations were made and the sample standard deviation was measured as 4. Calculate a 95% confidence interval for σ^2 of the form

$$(0, \text{upper}).$$

You may assume that the observations were randomly sampled from a normal distribution. The following R commands and output may be used.

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```

qchisq(0.05, 19)
## [1] 10.11701
qchisq(0.05, 19, lower.tail = FALSE)
## [1] 30.14353
qchisq(0.025, 19)
## [1] 8.906516
qchisq(0.025, 19, lower.tail = FALSE)
## [1] 32.85233

```

- (c) Prove that, even though S^2 is unbiased for σ , S is not unbiased for σ .

Hint: You may assume that $\text{var}(S) > 0$.

[9 marks]

3. Consider the multiple regression model

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\mathbf{Y}$ is an $n \times 1$ vector of response random variables, X is an $n \times p$ design matrix, $\boldsymbol{\beta}$ is a $p \times 1$ vector of regression parameters and $\boldsymbol{\epsilon}$ is an $n \times 1$ vector of random errors with $\epsilon_i \sim N(0, \sigma^2)$, $i = 1, \dots, n$.

- (a) Prove that if $X_{n \times p}$ is a matrix with linearly independent columns then the symmetric, $p \times p$ matrix $X^T X$ is invertible.
- (b) Prove that

$$(\mathbf{y} - X\hat{\boldsymbol{\beta}})^T (X\hat{\boldsymbol{\beta}} - X\boldsymbol{\beta}) = 0,$$

where

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}.$$

You may assume that the columns of X are linearly independent.

- (c) Hence, prove that

$$\|\mathbf{y} - X\boldsymbol{\beta}\|^2 = \|\mathbf{y} - X\hat{\boldsymbol{\beta}}\|^2 + \|X\hat{\boldsymbol{\beta}} - X\boldsymbol{\beta}\|^2$$

- (d) Hence, show that least squares estimates are given by

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}.$$

[16 marks]

4. Suppose $y_1, y_2, \dots, y_n$ are independent exponential observations with parameter λ , $\lambda > 0$. That is, for $i = 1, 2, \dots, n$,

$$f(y_i; \lambda) = \lambda e^{-\lambda y_i}, y_i > 0.$$

Please turn over for page 4

- (a) Write down the likelihood.
- (b) Write down the log-likelihood.
- (c) Find the maximum likelihood estimate of λ , $\hat{\lambda}$.
- (d) Find the Fisher information.
- (e) The following observations were made from an exponential distribution:

0.1, 0.2, 0.3, 0.5, 0.6, 0.1, 0.2, 0.2, 0.3, 0.2.

Calculate a 95% confidence interval for λ . You may assume that $P(Z < 1.96) = 0.975$, where $Z \sim N(0, 1)$.

- (f) An alternative form of the exponential distribution is

$$f(y_i; \beta) = \frac{1}{\beta} e^{-y_i/\beta}, y_i > 0, \beta > 0.$$

Give an expression for the maximum likelihood estimate of β .

[15 marks]

5. An analysis of the effect of displacement (`displ`) and class (`class`) on the highway fuel efficiency (`hwy`) for 38 popular models of car was performed in R. The commands and output are given in Appendix A.

The displacement of a car is volume of the cylinders, while the class is the type of car, in this case, we have just two levels - midsize and SUV. Of note, is the fact that the minimum displacement of SUV's is 2.5 litres.

- (a) Consider the scatterplot of highway fuel efficiency against displacement given in Figure 1. Describe the relationship.
- (b) Consider the separate regression model. Write down the two lines of best fit for the relationship between displacement and highway fuel efficiency: one for midsize cars and one for SUV cars.
- (c) Test for a statistically significant interaction term in the separate regression model at the 5% significance level. Remember to include the null and alternative hypotheses, the value of the test statistic, the P-value and your conclusion.
- (d) Calculate a 95% confidence interval for the slope in the identical model. The following R command may be useful. Interpret the confidence interval in context.

```
qt(0.975, 101)
## [1] 1.983731
```

- (e) Assess the assumptions of the linear model used in the parallel model. The plots given in Figure 2 may be used where appropriate.

[19 marks]

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Appendix A

```
## load libraries ----
library(tidyverse)

## Switch off significant stars - sorry folks - said I would ----
options(show.signif.stars=FALSE)

## Load MPG datasets ----
data(mpg)

## Filter for just midsize and SUV cars ----
mpg <- mpg %>%
  filter(class %in% c("midsize", "suv"))

## Look at relationship between fuel efficiency and displacement
ggplot(mpg, aes(x = displ, hwy, col = class)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(x = "Displacement (litres)",
       y = "Highway fuel efficiency (miles per gallon)") +
  theme(legend.position = "top")
```

```
## Identical regression model ----
identical.model <- lm(hwy ~ displ, data = mpg)
summary(identical.model)

##
## Call:
## lm(formula = hwy ~ displ, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.8605 -1.8725  0.1395  2.3221  8.1874
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   34.9028     1.0780   32.38  <2e-16
## displ        -3.4132     0.2676  -12.75  <2e-16
##
## Residual standard error: 3.256 on 101 degrees of freedom
## Multiple R-squared:  0.6169, Adjusted R-squared:  0.6131
## F-statistic: 162.7 on 1 and 101 DF, p-value: < 2.2e-16
```

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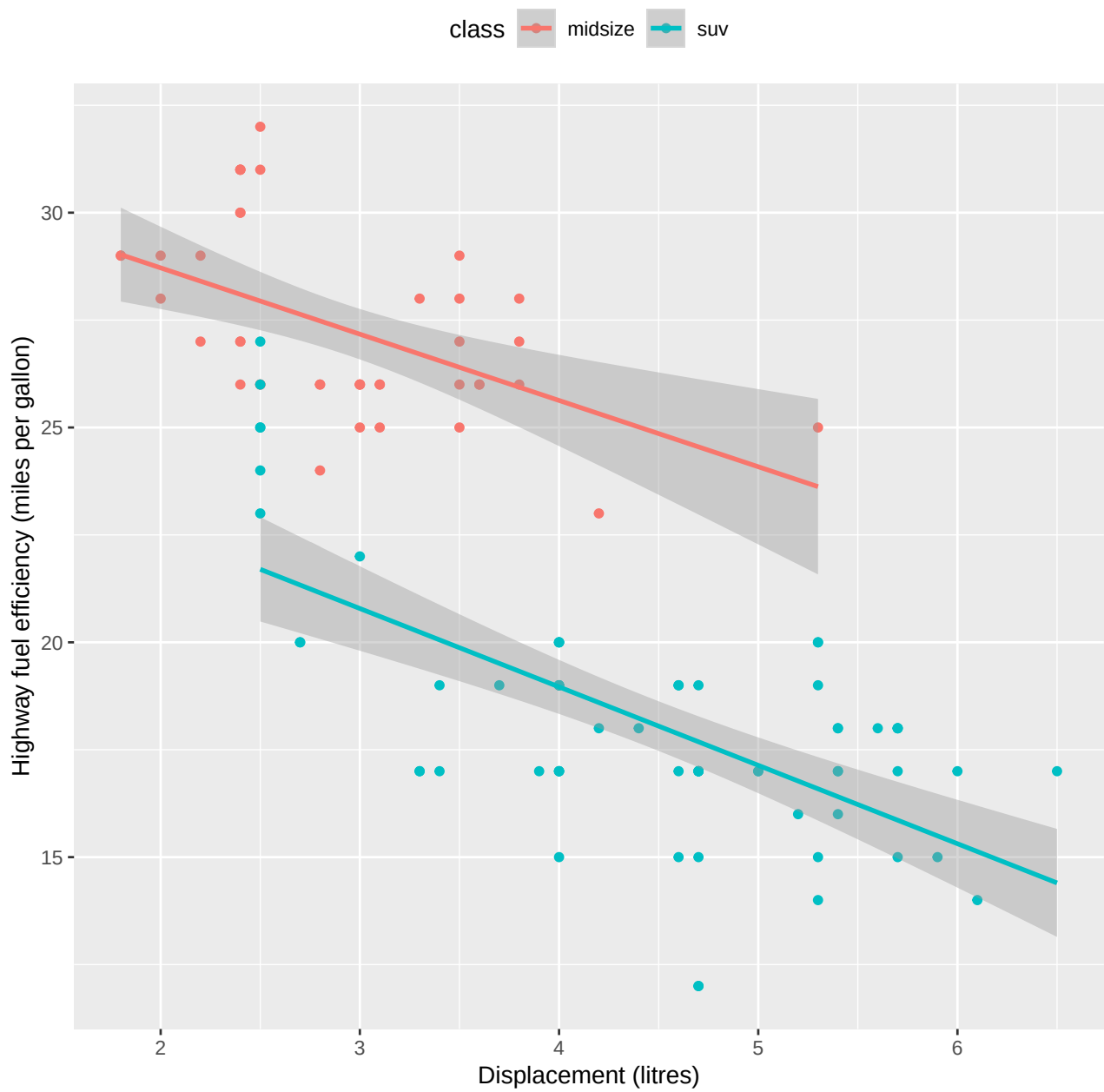


Figure 1: Scatterplot of highway fuel efficiency against displacement for midsize and SUV cars in MPG dataset.

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```
## Parallel regression model ----
parallel.model <- lm(hwy ~ displ + class, data = mpg)
summary(parallel.model)

##
## Call:
## lm(formula = hwy ~ displ + class, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.7003 -1.2284 -0.2284  1.5318  5.4273
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  32.4358     0.7285  44.524 < 2e-16
## displ       -1.7602     0.2224  -7.915 3.46e-12
## classssuv    -6.4627     0.5447 -11.865 < 2e-16
##
## Residual standard error: 2.109 on 100 degrees of freedom
## Multiple R-squared:  0.8409, Adjusted R-squared:  0.8377
## F-statistic: 264.3 on 2 and 100 DF,  p-value: < 2.2e-16
```

```
## Separate regression model ----
separate.model <- lm(hwy ~ displ * class, data = mpg)
summary(separate.model)

##
## Call:
## lm(formula = hwy ~ displ * class, data = mpg)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.6846 -1.3344 -0.2321  1.4848  5.3006
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   31.8013     1.4006  22.706 < 2e-16
## displ        -1.5430     0.4658  -3.313  0.00129
## classssuv     -5.5397     1.8215  -3.041  0.00302
## displ:classssuv -0.2819     0.5307  -0.531  0.59646
##
## Residual standard error: 2.117 on 99 degrees of freedom
## Multiple R-squared:  0.8414, Adjusted R-squared:  0.8366
```

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```
## F-statistic: 175 on 3 and 99 DF, p-value: < 2.2e-16
```

```
## Plots for assumption checking ----  
tmp <- par(mfrow = c(2,2))  
plot(parallel.model)  
par(tmp)
```

Please turn over for page 9

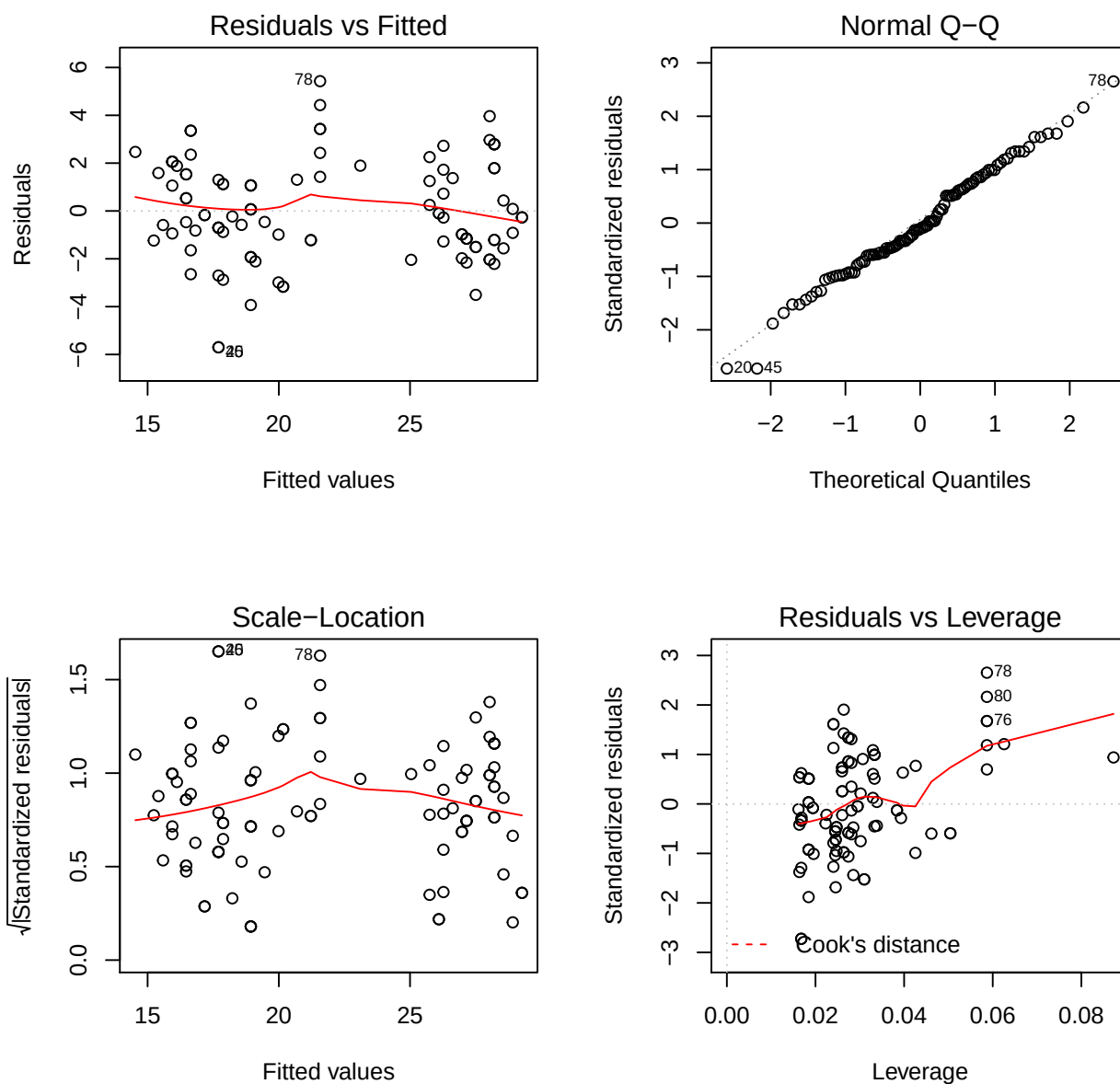


Figure 2: Plots to check assumptions for the parallel regression model

Please turn over for page 10

Appendix B

Binomial Distribution

- $p(x) = \binom{n}{x} p^x (1-p)^{n-x}$ for $x = 0, 1, 2, \dots, n$
- $E(X) = np$
- $\text{var}(X) = np(1-p)$

Geometric Distribution

- $p(x) = p(1-p)^{x-1}$ for $x = 1, 2, \dots$
- $E(X) = \frac{1}{p}$
- $\text{var}(X) = \frac{1-p}{p^2}$

Poisson Distribution

- $p(x) = \frac{e^{-\lambda} \lambda^x}{x!}$ for $x = 0, 1, 2, \dots$
- $E(X) = \lambda$
- $\text{var}(X) = \lambda$

Uniform Distribution

- $f(x) = \frac{1}{b-a}$ for $a < x < b$
- $E(X) = \frac{a+b}{2}$
- $\text{var}(X) = \frac{(b-a)^2}{12}$

Exponential Distribution

- $f(x) = \lambda e^{-\lambda x}$ for $x > 0$
- $E(X) = \frac{1}{\lambda}$
- $\text{var}(X) = \frac{1}{\lambda^2}$

Gamma Distribution

- $f(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ for $x > 0$
- $E(X) = \frac{\alpha}{\lambda}$
- $\text{var}(X) = \frac{\alpha}{\lambda^2}$

Normal Distribution

- $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(1/2\sigma^2)(x-\mu)^2}$ for $-\infty < x < \infty$
- $E(X) = \mu$
- $\text{var}(X) = \sigma^2$